Jung-hun Kim

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Education

Korea Advanced Institute of Science and Technology (KAIST) Aug 2018–Aug 2023
Ph.D. in Industrial & Systems Engineering, GPA: 3.94/4.30
Topic: Online learning and decision-making under nonstationary environments
Advisor: Se-Young Yun

Korea Advanced Institute of Science and Technology (KAIST) Sep 2016–Aug 2018
M.S. in Graduate School of Knowledge Service Engineering, GPA: 4.03/4.30

Ulsan National Institute of Science and Technology (UNIST) Mar 2011–Aug 2015
B.S. in Physics (Minor: Mechanical Engineering), *Summa Cum Laude*

Research Experience

CREST, ENSAE, Institut Polytechnique de Paris – FairPlay Joint Team Palaiseau, France
Postdoctoral Researcher (Supervisor: Vianney Perchet) Jun 2025 - Present

- Online learning for optimal stopping and sequential decision-making under uncertainty

Seoul National University Seoul, South Korea
Postdoctoral Researcher (Supervisor: Min-hwan Oh) Sep 2023 - May 2025

- Learning-based control for queueing, matching, and combinatorial stochastic systems

London School of Economics, Department of Statistics London, UK
Research Intern (Supervisor: Milan Vojnović) Mar 2021 - Jul 2021

- Learning-based scheduling and resource allocation in parallel-server queues

Research Interests

Online learning, stochastic optimization, and data-driven decision-making under uncertainty and non-stationarity, with applications to industrial and service systems, including scheduling, queueing, matching, resource allocation, pricing, and optimal stopping.

Publications

1. [**NeurIPS 2026, Accepted**] Stochastic Matching Bandits with Rare Optimization Updates (Arxiv);
Jung-hun Kim and Min-hwan Oh
2. [**IEEE Transactions on Networking 2026**] Learning to Schedule in Parallel-Server Queues with Stochastic Bilinear Rewards (Arxiv, IEEE Xplore)
Jung-hun Kim and Milan Vojnović
3. [**ICLR 2026**] Learning in Prophet Inequalities with Noisy Observations (Arxiv, OpenReview)
Jung-hun Kim, Vianney Perchet
4. [**NeurIPS 2025**] Oracle-Efficient Combinatorial Semi-Bandits (Arxiv, OpenReview)
Jung-hun Kim, Milan Vojnović, Min-hwan Oh

5. [**TMLR 2025**] Adversarial Bandits Against Arbitrary Strategies (Arxiv, OpenReview)
Jung-hun Kim, Se-Young Yun.

6. [**ICML 2025**] Tracking Most Significant Shifts in Infinite-Armed Bandits (Arxiv, OpenReview)
Joe Suk, **Jung-hun Kim**.

7. [**ICLR 2025**] Dynamic Assortment Selection and Pricing with Censored Preference Feedback (Arxiv, OpenReview)
Jung-hun Kim, Min-hwan Oh.

8. [**NeurIPS 2024**] Queueing Matching Bandits with Preference Feedback (Arxiv, Proceedings)
Jung-hun Kim, Min-hwan Oh

9. [**NeurIPS 2024**] An Adaptive Approach for Infinitely Many-armed Bandits under Generalized Rotting Constraints (Arxiv, OpenReview)
Jung-hun Kim, Milan Vojnović, Se-Young Yun

10. [**AISTATS 2023**] Contextual Linear Bandits under Noisy Features: Towards Bayesian Oracles (Arxiv, Proceedings)
Jung-hun Kim, Se-Young Yun, Minchan Jeong, Jun Hyun Nam, Jinwoo Shin, Richard Combes

11. [**ICML 2022**] Rotting Infinitely Many-armed Bandits (Arxiv, Proceedings)
Jung-hun Kim, Milan Vojnović, Se-Young Yun

12. [**IEEE Bigdata 2018**] Research Hypothesis Generation Using Link Prediction in a Bipartite Graph (Proceedings)
Jung-hun Kim, Aviv Segev

Working Papers

1. Prophet Inequalities with Discounted Rewards (Arxiv); **Jung-hun Kim** and Vianney Perchet

2. Asymptotically Optimal Learning in Parametric Prophet Inequalities (Arxiv);
Jung-hun Kim, Anna Grebennikova, and Vianney Perchet

Teaching

- **Tutorial Instructor** at ENSAE Paris
  Theoretical Foundations of Machine Learning (APM_4ML04_AE) — Spring 2026
  Advanced Machine Learning (APM_5ML09_AE) — Fall 2025
- **Teaching Assistant** at KAIST
  Statistical Machine Learning (IE343) — Spring 2019, 2020
  Concentration inequalities in matrix data analysis (IE801(B)) — Fall 2018
  General Physics I (PH141) — Fall 2015
- **Teaching Tutorials** at Enterprise
  Samsung DS-KAIST AI Expert Program — Summer, Fall 2019
  LG-KAIST AI & Big Data Advanced Course — Spring 2019

Academic Service

Area Chair
ICML, ICLR

Reviewer
NeurIPS, ICML, ICLR, AISTATS, JMLR, CDC, AAAI

Presentations

CentraleSupélec R3 Research Group Seminar — Gif-sur-Yvette, France (remote)
– *Learning and Control in Stochastic Decision Systems* — Oct 2026
Queueing, scheduling, and combinatorial learning

KAIST ISE/GSDS Seminar — Daejeon, South Korea
– *Learning and Control in Stochastic Decision Systems* — Aug 2026
Queueing, Matching, and Combinatorial Learning

Yonsei University Department of Applied Statistics Seminar — Seoul, South Korea
– *Learning and Optimal Stopping* — Aug 2026
Sequential Decision-Making under Noisy and Nonstationary Information

International Laboratory on Learning Systems (ILLS) Seminar — Montréal, Canada (remote)
– *I.i.d. Prophet Inequalities with Discounted Rewards* — Jul 2026

Seoul National University Department of Statistics Seminar — Seoul, South Korea
– *Reliable Learning and Decision-Making under Uncertainty* — May 2026
Online Learning, Optimal Stopping, and Stochastic Systems

FairPlay Seminar — Criteo, Paris, France
– *Combinatorial Bandit Problems Motivated by Applications* — Sep 2025

The Institute for Data Innovation in Science (IDIS) Seminar — SNU, Seoul, South Korea
– *Queueing Matching Bandits with Preference Feedback* — Dec 2024

INFORMS Annual Meeting – APS Showcase — Seattle, Washington, U.S.
– *Sequential Decision-Making under Preference Feedback* — Oct 22, 2024
APS PhD Student Showcase 1 (flash talk and poster)

INFORMS Annual Meeting — Seattle, Washington, U.S.
– *Queueing Matching Bandits with Preference Feedback* — Oct 20, 2024
Invited talk: Learning in Sharing Economy and E-Commerce

SIAM Conference on Optimization (OP23) — Seattle, Washington, U.S.
– *Scheduling Servers with Stochastic Bilinear Rewards* — Jun 2023

Projects

- Design of Optimal Online Matching Markets — 2025–2027
(funded by French National Research Agency (ANR), DOOM project)
- **Multi-Objective Optimization and Fairness of Exposure in Bandit Algorithms** — 2023–2025
(**Principal Investigator**; funded by the Ministry of Education, Republic of Korea, G-LAMP Program)
Total grant amount: KRW 34 million
- Fast Adaptive Prediction Model for Sparse Big Data — 2022–2023
(funded by Samsung Research)
- Scheduling Jobs with Complex Delay Costs in Data Processing Platforms — 2020–2021
(research project at the London School of Economics and Political Science;
funded by Meta (formerly Facebook), Systems for Machine Learning Research Award)
- Development of Prediction Models for Tire Properties and Inverse Design Models for Tire Recipes — 2019–2020
(funded by Hankook Tire & Technology Group)

Fellowships, Scholarships, and Awards

- **NRF Postdoctoral Fellowship Abroad**, National Research Foundation of Korea (NRF), Korean Government 2026–2027
  Project: *Learning for Optimal Stopping in Nonstationary Environments*
  Grant amount: KRW 60 million (approx. EUR 36,700)
- The National Scholarship for Science and Engineering, Korea Student Aid Foundation 2013–2014
- Award for Semester Academic Excellence, UNIST 5 times in 2011–2014
- Scholarship for Academic Excellence, UNIST 2011–2012

Extra

- Master Student in Physics, KAIST Fall 2015–Spring 2016

References

- Prof. Vianney Perchet, CREST, ENSAE, IP Paris
  Email: vianney.perchet@normalesup.org
- Prof. Se-Young Yun, KAIST
  Email: yunseyoung@kaist.ac.kr
- Prof. Min-hwan Oh, Seoul National University
  Email: minoh@snu.ac.kr
- Prof. Milan Vojnović, London School of Economics
  Email: m.vojnovic@lse.ac.uk